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(54) **FENESTRATION UNIT WITH ADJUSTABLE BOLT RECEIVER**

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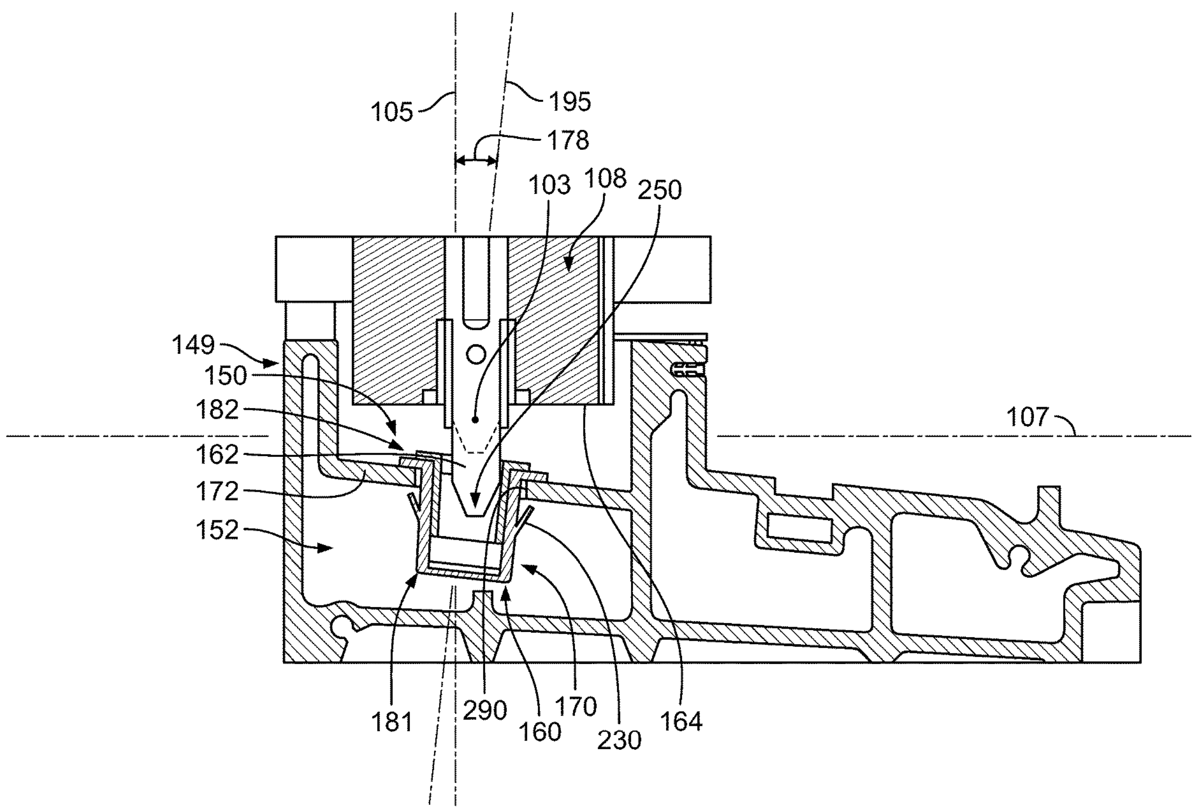
(57) **ABSTRACT**

An adjustable bolt receiver includes a first member and a second member. The first member includes at least one first engagement feature. The second member includes a strike head and an insert projection that projects from the strike head. The insert projection is receivable in a first opening of the first member in a first position along the axis and, alternatively, in a second position along the axis. The second member includes at least one second engagement feature configured to engage with the at least one first engagement feature for retaining the insert projection in the first position and in the second position along the axis. The insert projection at least partly defines a second opening configured to receive a bolt when in the extended position to secure the panel to the panel support.



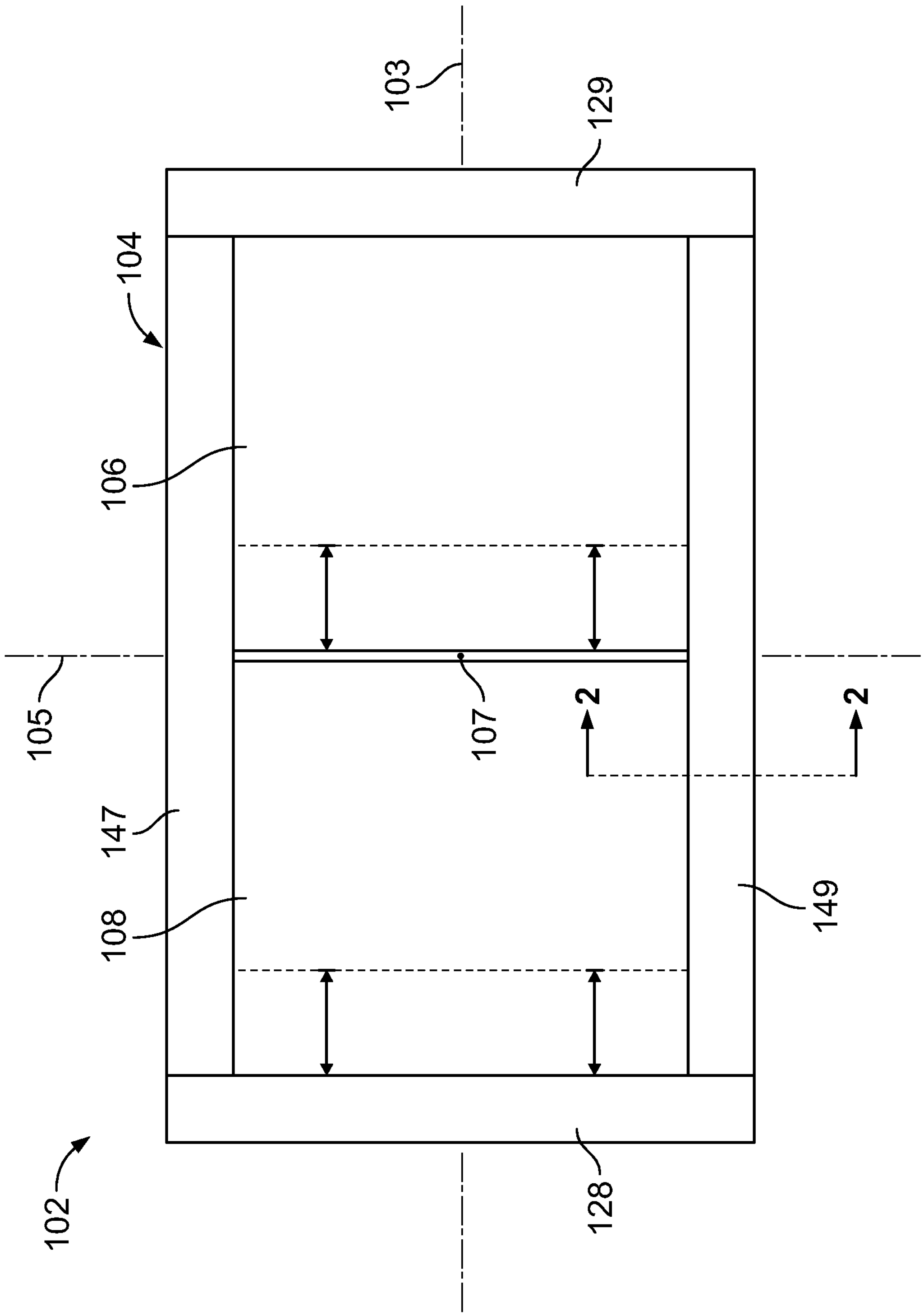
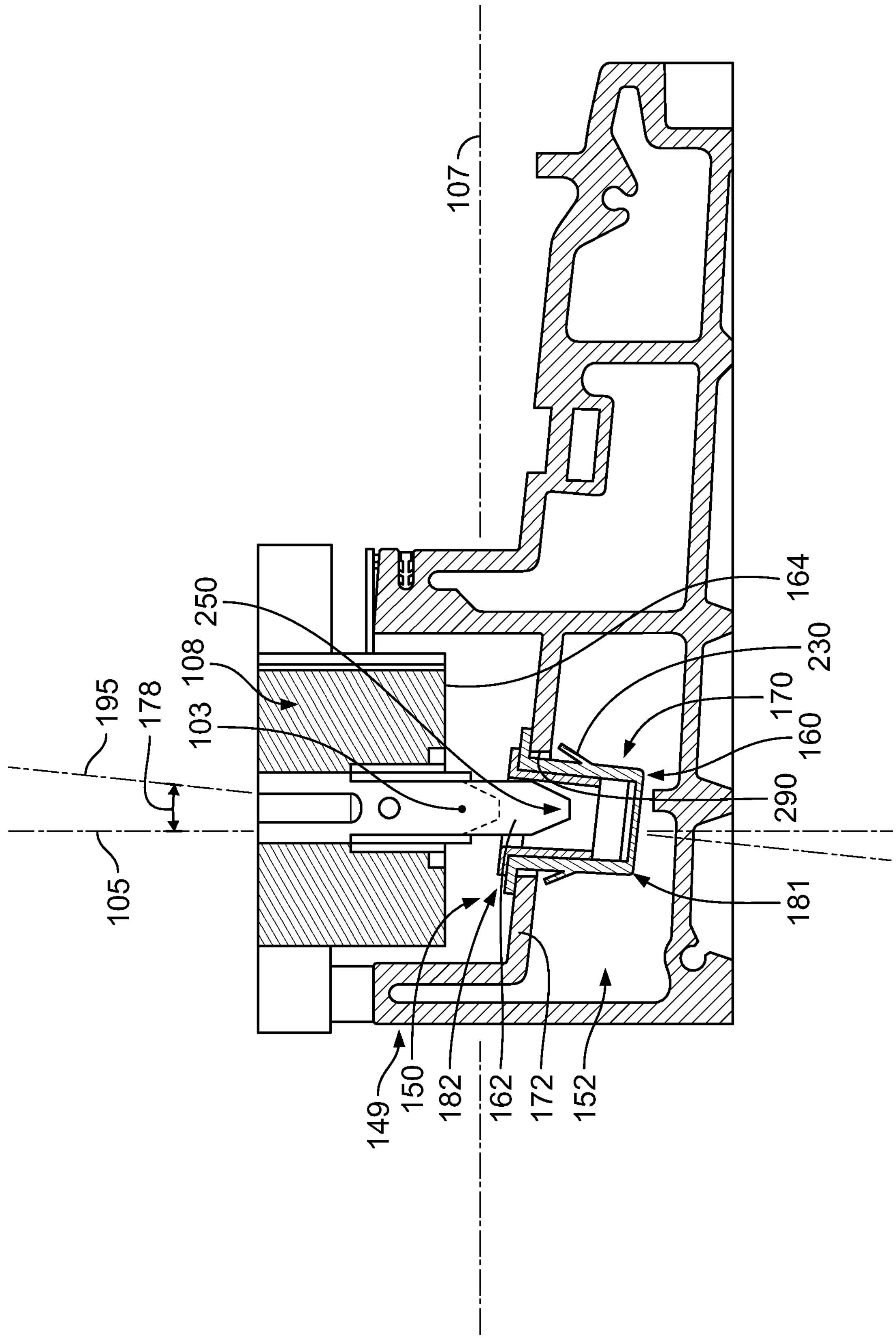


FIG. 1


107
250
108
103
178
195
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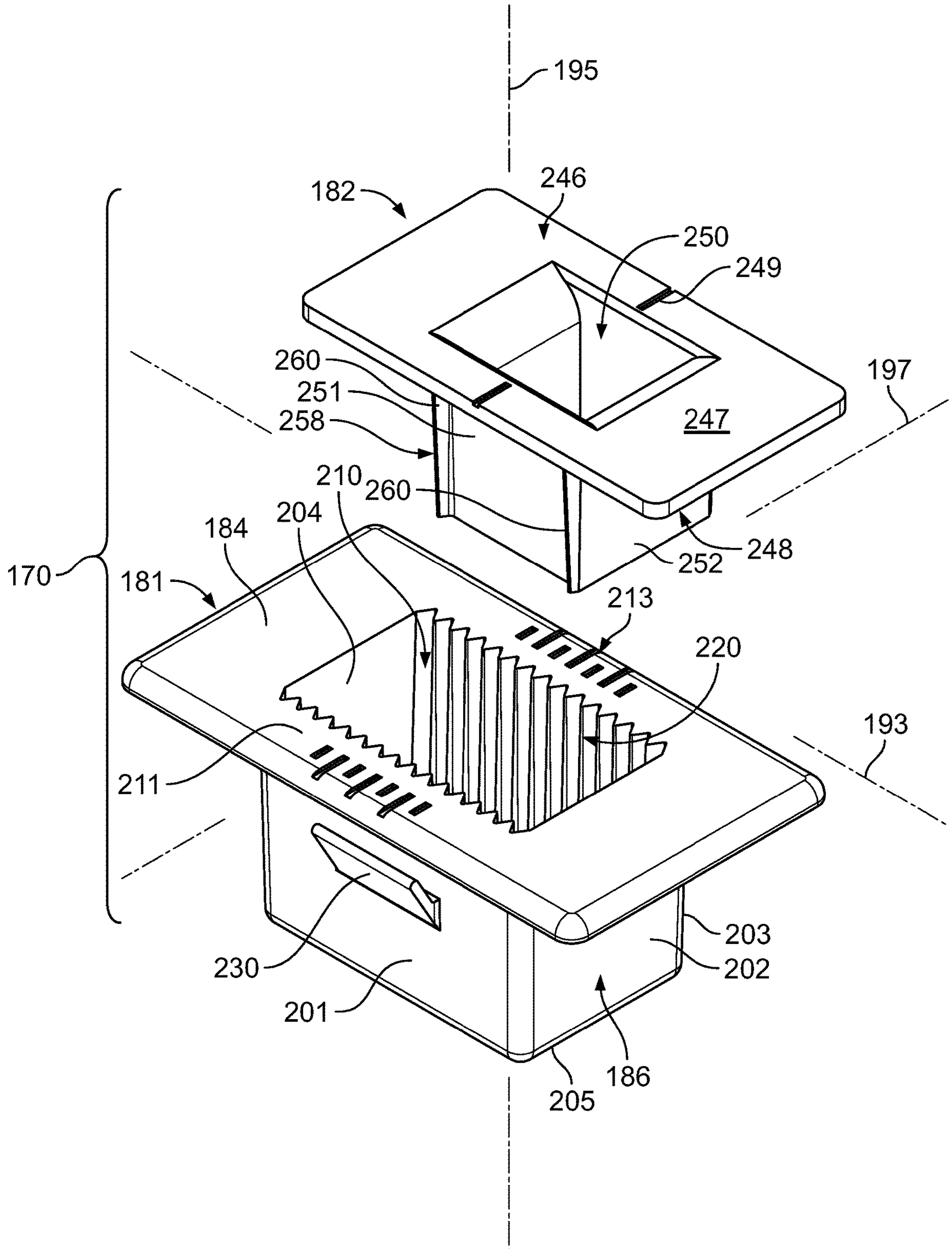
162
172
152

FIG. 2


195
182
246
250
249
260
251
258
197
247
210
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252
170
181
213
220
193
211
230
203
202
201
186
205

FIG. 3

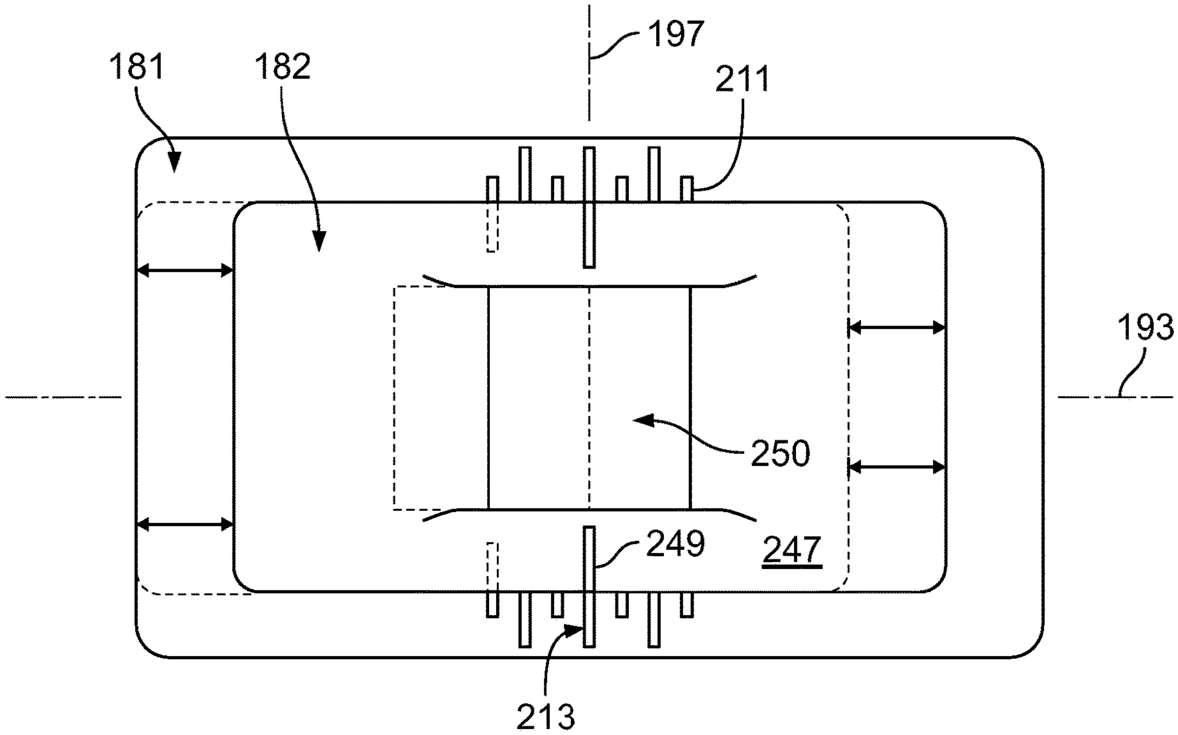

197
181
182
211
193
250
249
247
213

FIG. 4

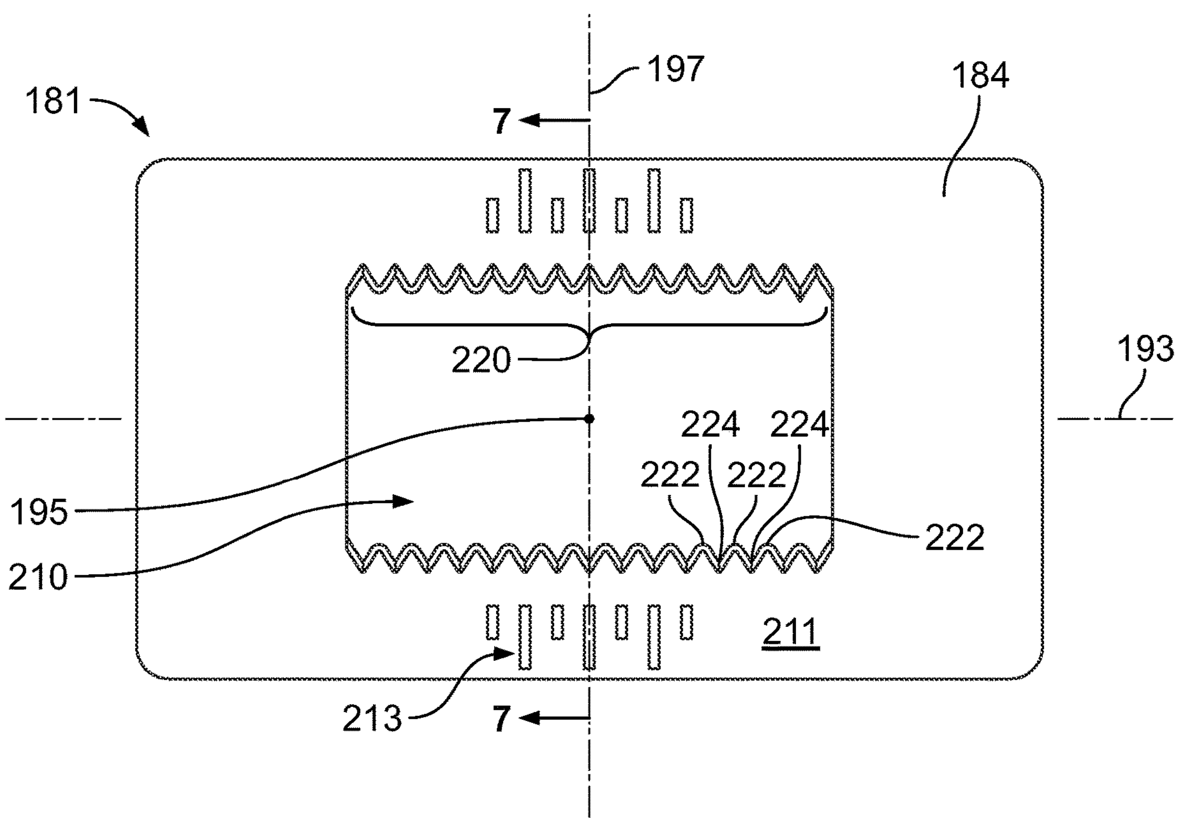

197
181
7
184
220
193
224
224
222
222
195
222
210
211
213
7

FIG. 5

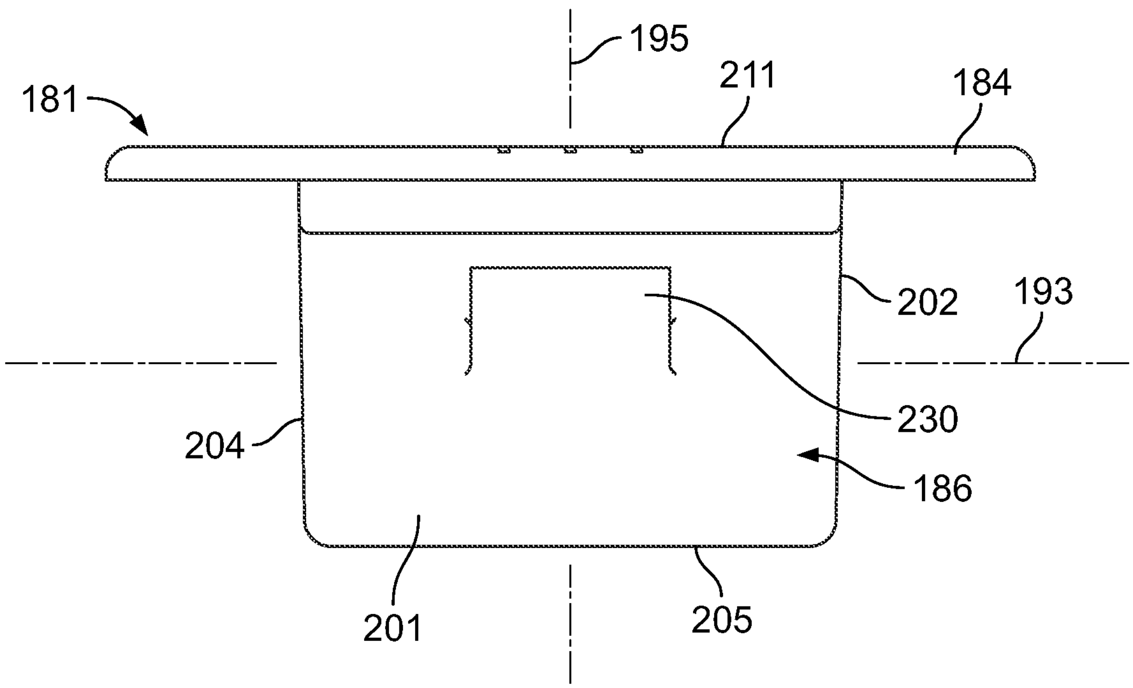

195
211
184
181
202
193
230
204
186
201
205

FIG. 6

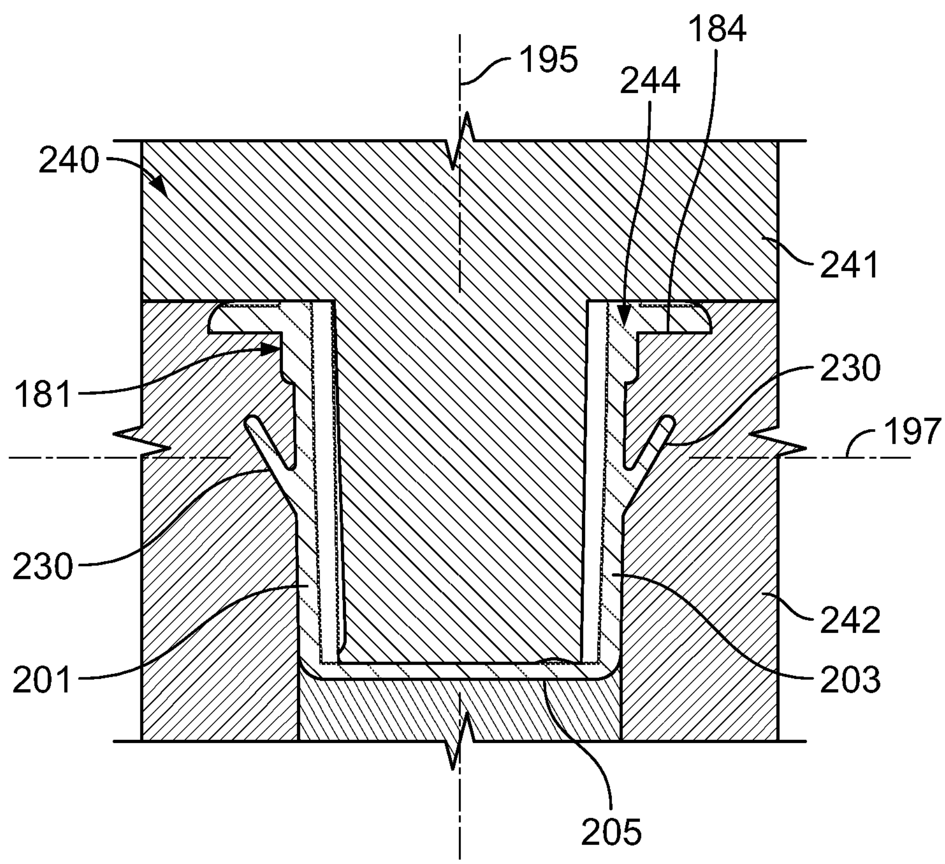
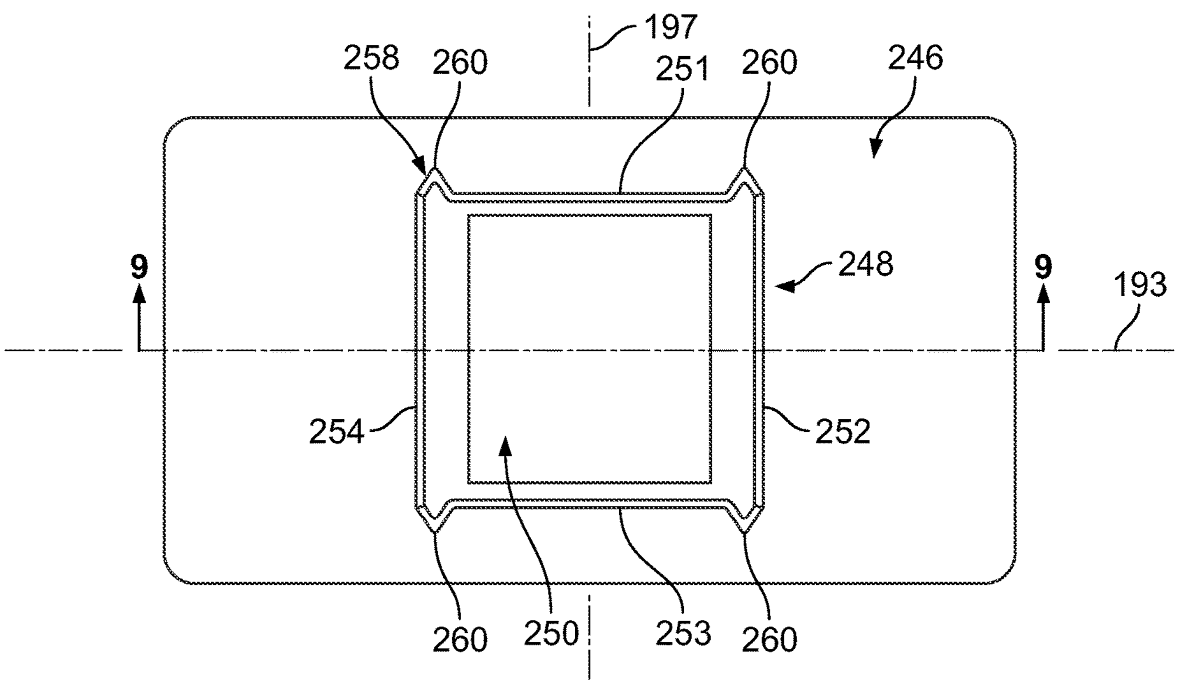

184
195
244
240
241
181
230
197
230
242
201
203
205

FIG. 7


197
258
260
251
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246
9
248
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193
254
252
260
250
253
260

FIG. 8

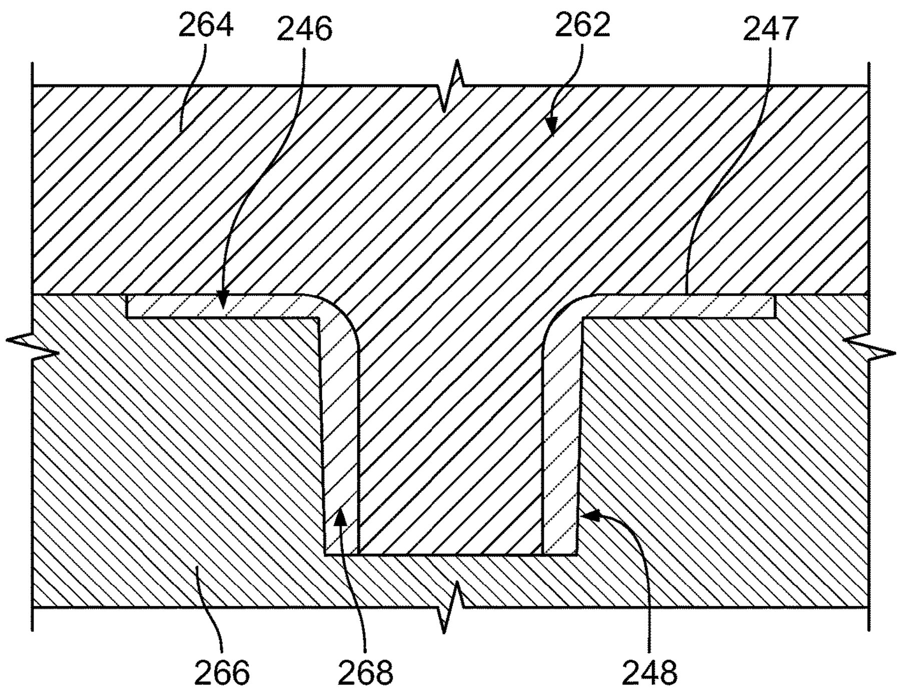
264
246
262
247
266
268
248

FIG. 9

FENESTRATION UNIT WITH ADJUSTABLE BOLT RECEIVER

TECHNICAL FIELD

[0001] The field of this disclosure relates generally to fenestration units and, more particularly, to a fenestration unit with an adjustable bolt receiver.

BACKGROUND

[0002] Fenestration units, such as sliding patio door units, sliding window units, etc. may include a bolt securement system. For example, a panel (e.g., door leaf, window sash, etc.) of the fenestration unit may be supported by a support structure (e.g., frame, sill, foundation, header, etc.), and the bolt securement system may include a bolt that is moveably attached to the panel and a receiver hole in the support structure. The bolt (e.g., a shoot-bolt, flush-bolt, etc.) may be selectively extended outward from the panel and into the receiver hole to secure the panel to the support structure.

[0003] However, it can be difficult, time consuming, and costly to ensure proper alignment between the sliding axis of the bolt and the receiving hole. Some bolt securement systems may be undesireably bulky or otherwise visually displeasing. Furthermore, the bolt securement system may be difficult, expensive, and/or time consuming to manufacture. Installation and/or repair of the bolt adjustment system may also be difficult and time consuming. The part count may be relatively high, which may disadvantageously increase complexity and cost.

[0004] Accordingly, there is a need for a bolt securement system for a fenestration unit for more convenient and ergonomic alignment of the bolt and receiving member. Furthermore, there is a need for a bolt securement system that is compact and aesthetically pleasing. There is also a need for a bolt securement system that may be manufactured inexpensively, with a relatively low part count, etc. Additional aspects and benefits will be apparent from the following detailed description of example embodiments, which proceeds with reference to the accompanying drawings.

BRIEF SUMMARY

[0005] In an example embodiment, an adjustable bolt receiver for selectively securing a panel of a fenestration unit to a panel support of the fenestration unit is disclosed wherein the fenestration unit has a bolt that is slidably attached to the panel for movement between a retracted position and an extended position relative to the panel for selectively engaging with the panel support. The adjustable bolt receiver includes a first member configured to be fixedly attached to the panel support, and the first member has a first opening defining an axis. The first member includes at least one first engagement feature. Moreover, the adjustable bolt receiver includes a second member that includes a strike head and an insert projection that projects from the strike head. The insert projection is receivable in the first opening in a first position along the axis and, alternatively, in a second position along the axis. The second member includes at least one second engagement feature configured to engage with the at least one first engagement feature for retaining the insert projection in the first position and in the second position along the axis. The insert projection at least partly defines a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support.

[0006] Also, in another embodiment, a method of manufacturing an adjustable bolt receiver configured for selectively securing a panel of a fenestration unit to a panel support of the fenestration unit is disclosed, wherein the fenestration unit has a bolt that is slidably attached to the panel for movement between a retracted position and an extended position relative to the panel for selectively engaging with the panel support. The method includes providing a first member configured to be fixedly attached to the panel support, and the first member has a first opening defining an axis. The first member includes at least one first engagement feature. Furthermore, the method includes providing a second member that includes a strike head and an insert projection that projects from the strike head. The second member includes at least one second engagement feature, and the insert projection at least partly defining a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support. The method also includes selecting one of a first position for the insert projection in the first opening along the axis and a second position for the insert projection in the second opening along the axis. Additionally, the method includes inserting the insert projection in the selected one of the first position and the second position. The at least one second engagement feature is configured to engage with the at least one first engagement feature for retaining the insert projection in the first position and, alternatively, in the second position along the axis.

[0007] Additionally, in a further embodiment, a fenestration unit is disclosed that includes a panel with a bolt that is slidably attached thereto for movement between a retracted position an extended position relative to the panel. The fenestration unit includes a frame that supports the panel for sliding movement within the frame along an axis, and the frame includes a sill. Moreover, the fenestration unit includes an adjustable bolt receiver that includes a first member configured to be fixedly attached to the sill, and the first member has a first opening that extends along the axis. The first member includes at least one array of ridges and slots. Also, the adjustable bolt receiver includes a second member that includes a strike head and an insert projection that projects from the strike head. The insert projection is receivable in the first opening in a first position along the axis and, alternatively, in a second position along the axis. The second member includes at least one flange that is removably engaged within one of the slots of the array in the first position and in another of the slots of the array in the second position. The insert projection at least partly defines a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support.

BRIEF DESCRIPTION OF DRAWINGS

[0008] The various embodiments will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:

[0009] FIG. **1** is a schematic plan view of a fenestration unit with a bolt securement system according to example embodiments of the present disclosure;

[0010] FIG. **2** is a cross-sectional view of the fenestration unit and bolt securement system taken along the line **2-2** of FIG. **1**;

[0011] FIG. **3** is an exploded perspective view of an adjustable bolt receiver of the bolt securement system of FIG. **2**;

[0012] FIG. **4** is a plan view of the adjustable bolt receiver of FIG. **3**;

[0013] FIG. **5** is a plan view of a first member of the adjustable bolt receiver of FIG. **3**;

[0014] FIG. **6** is a side view of the first member of FIG. **5**;

[0015] FIG. **7** is a cross-sectional view of the first member taken along the line **7-7** of FIG. **5**, wherein the first member is shown in a mold cavity during an exemplary manufacturing method of the adjustable bolt receiver;

[0016] FIG. **8** is a bottom view of a second member of the adjustable bolt receiver of FIG. **3**; and

[0017] FIG. **9** is a cross-sectional view of the second member taken along the line **9-9** of FIG. **8**, wherein the second member is shown in a mold cavity during an exemplary manufacturing method of the adjustable bolt receiver.

DETAILED DESCRIPTION

[0018] The following Detailed Description is merely exemplary in nature and is not intended to limit the various embodiments or the application and uses thereof. Furthermore, there is no intention to be bound by any theory presented in the preceding background or the following detailed description.

[0019] Generally, embodiments of the present disclosure include a fenestration unit with a bolt securement system. The bolt securement system may include a bolt that may be selectively extended from a panel of the fenestration unit. The bolt securement system may also include an adjustable bolt receiver configured to receive the bolt. The position of the adjustable bolt receiver may be selectively adjusted for conveniently aligning with the sliding axis of the bolt. Methods of manufacturing and assembling the fenestration unit, bolt securement system, and adjustable bolt receiver are also disclosed.

[0020] The adjustable bolt receiver may include a first member that may be attached, for example, to the sill of a fenestration frame. The first member may, in some embodiments, be snap-fit into the sill. The adjustable bolt receiver may further include a second member that may be fixedly attached to the first member in one of a plurality of positions.

[0021] For example, the second member may include a strike head and an insert projection with an opening that extends through the second member and through the insert projection. The opening may be sized to receive the bolt when extended out from the panel. The insert projection may be inserted into a larger opening defined in the first member in one of a plurality of positions. The first and second members may include complimentary engagement features, such as tapered flanges of one of the members that are received in respective slots of the other. Moreover, in some embodiments, the first member and/or the second member may have symmetric features.

[0022] Accordingly, aligning the bolt and the bolt receiver may be selectable and adjustable for high convenience. The second member may also be strong and robustly secured within the first member once the second member has been positioned.

[0023] Also, the adjustable bolt receiver may have a relatively low part count for added manufacturing efficiency. The adjustable bolt receiver may be highly manufacturable and may be packaged neatly. The first member may nest within the second member and both members may be received, for example, within the sill; therefore, the bolt receiver may be compact, may have a low-profile, and may be aesthetically pleasing. Manufacture, assembly, and use may also be convenient and less time consuming.

[0024] FIG. **1** illustrates a fenestration unit **102** according to example embodiments of the present disclosure. The fenestration unit **102** may be configured as a door unit in some embodiments. In some embodiments, the fenestration unit **102** may be configured as a slider door fenestration unit **102**, such as with horizontally-sliding patio door unit for a building, dwelling, or other structure. However, it will be appreciated that features of the present disclosure may be included in hingedly-attached door panel system, a window fenestration unit or another fenestration unit without departing from the scope of the present disclosure.

[0025] The fenestration unit **102** may include a support structure, such as a rectangular frame **104** that defines a first axis **103**, a second axis **105**, and a third axis **107**, which are orthogonal to each other. The first axis **103** may be a horizontal axis extending between a first jamb **128** and a second jamb **129** of the frame **104**. The second axis **105** may be a vertical axis that is generally aligned with the direction of gravity and that extends between a header **147** and a sill **149** of the frame **104**. The third axis **107** may be a horizontal axis extending in an interior-exterior direction through the frame **104**.

[0026] The fenestration unit **102** may also include at least one panel, such as a first panel **106**, and a second panel **108** that are supported by the frame **104**. The panels **106**, **108** may comprise door panels (i.e., door leafs) in some embodiments. In other embodiments, the panels **106**, **108** may comprise window sashes. Furthermore, the panels **106**, **108** may include and support glazing units (e.g., those with one or more panes of glass) in some embodiments.

[0027] One or both of the first and second panels **106**, **108** may be moveably supported within the frame **104**. In some embodiments, the first panel **106** may be fixed within the frame **104**, and the second panel **108** may be supported for movement within the frame **104** between a closed position (shown in solid lines) and an open position (shown in phantom). In some embodiments, the first axis **103** defined by the frame **104** may be an axis for sliding the second panel **108** open and closed. The second panel **108** may be supported for sliding movement within the frame **104** along the first axis **103** as the second panel **108** moves within the frame **104** between the closed and open positions.

[0028] As shown in FIG. **2**, the second panel **108** may be supported in a channel **150** of the sill **149**, and the second panel **108** may be supported for sliding movement along the axis **103** therein (e.g., by one or more rollers, etc.). As such, the second panel **108** may slidingly move relative to the first panel **106** when moving between the closed and open positions. In some embodiments, the sill **149** may include a plurality of walls that define a hollow interior. For example, the sill **149** may include a sill cavity **152**, which is spaced below the channel **150**.

[0029] Additionally, the fenestration unit **102** may include a bolt securement system **160** for selectively securing the panel **108** to the frame **104**. The bolt securement system **160** may include a bolt **162** (i.e., shank, etc.) that is included on the panel **108**. The bolt **162** may be mounted to the panel **108** so as to move along a sliding axis that may be substantially parallel to the second axis **105** (i.e., upward and downward).

The bolt **162** may be slidably attached to the panel **108** for movement between a retracted position (shown in phantom) and an extended position (shown in solid lines) relative to a bottom edge **164** of the panel **108**. The bolt **162** may be operatively coupled to a knob, handle, or other selector device with which a user may selectively move the bolt **162** between the retracted and extended positions. Thus, the bolt **162** may be selectively moved relative to the sill **149** for securing the panel **108** to the sill **149**; however, it will be appreciated that the bolt securement system **160** may be configured for securing the panel **108** to another member of the frame **104** without departing from the scope of the present disclosure.

[0030] The bolt securement system **160** may also include an adjustable bolt receiver **170**, which is configured to receive the bolt **162** in the extended position. The bolt receiver **170** may be attached, supported, and mounted to the sill **149**. For example, the bolt receiver **170** may be attached to and may extend through a top wall **172** of the sill **149** so as to extend partly into the sill cavity **152**. The adjustable bolt receiver **170** may also include an opening **250** that defines a receiver axis **195**. The receiver axis **195** may be disposed at an angle **178** relative to the sliding axis of the bolt (e.g., the second axis **105**). However, when aligned along the sliding axis of the panel **108** (i.e., the first axis **103**), there may be sufficient size tolerancing between the bolt **162** and the opening **250** to allow the bolt **162** to slide freely in and out of the opening **250**.

[0031] The bolt **162** may be selectively extended into the bolt receiver **170** so as to provide interference against sliding movement of the panel **108** along the axis **103**. The bolt **162** may be selectively retracted out of the bolt receiver **170** to thereby allow sliding movement of the panel **108** along the axis **103**.

[0032] Furthermore, as will be discussed, the bolt receiver **170** may be adjustable between multiple positions so as to selectively position the opening **250** relative to the sill **149**. Accordingly, the bolt receiver **170** may be selectively moved and adjusted for conveniently aligning with the bolt **162** (i.e., for conveniently aligning the receiver axis **195** and the sliding axis of the bolt **162** along the sliding axis of the panel **108**).

[0033] Referring now to FIG. **3**, the adjustable bolt receiver **170** is shown in additional detail. As shown, the adjustable bolt receiver **170** may generally include a first member **181** (i.e., a retainer member) and a second member **182** (i.e., a receiver member) that are removably attached together. For reference purposes, a receiver Cartesian coordinate system is shown, which includes a first receiver axis **193**, a second receiver axis **195**, and a third receiver axis **197**. As indicated in FIG. **2** and as will be discussed, the adjustable bolt receiver **170** may be attached to the sill **149** such that the first receiver axis **193** is aligned and parallel to the sliding axis of the panel **108** (i.e., aligned and parallel to the first axis **103**); however, the bolt receiver **170** may be canted slightly, for example, at the angle **178** mentioned above.

[0034] As shown in FIGS. **3**, **5**, **6**, and **7**, the first member **181** may be unitary, one-piece, monolithic part that is hollow and box-shaped. The first member **181** may be constructed of a rigid and strong polymeric material. In some embodiments, the first member **181** may be constructed of a material suitable for molding (e.g., injection molding). The first member **181** may include a support head **184** and a base extension **186**. the support head **184** may comprise a flat plate that lies substantially within a plane defined by the first and third receiver axes **193**, **197**. The base extension **186** may be rectangular with a first wall **201**, a second wall **202**, a third wall **203**, and a fourth wall **204** that extend along the second receiver axis **195** from the underside of the support head **184**. The base extension **186** may also include a bottom wall **205** that is attached to the first, second, third, and fourth walls **201**, **202**, **203**, **204** and that is substantially parallel to and spaced apart from the support head **184** along the second receiver axis **195**. The first member **181** may further include a first opening **210**. The first opening **210** may be a somewhat rectangular recess or blind hole that extends through the support head **184** and into the base extension **186**. Thus, the first receiver axis **193** may extend between the second and fourth walls **202**, **204**, and the third receiver axis **197** may extend between the first and third walls **201**, **203**. The second receiver axis **195** may be the longitudinal axis (hole axis) of the first opening **210**. The space between the second and fourth walls **202**, **204** (along the first receiver axis **193**) may be greater than the space between the first and third walls **201**, **203** (along the third receiver axis **197**); therefore, the first receiver axis **193** may be the major axis of the first opening **210**, and the third receiver axis **197** may be the minor axis of the first opening **210**.

[0035] Additionally, the first member **181** may further include a lip **211**. The lip **211** may be defined by the upper surface of the support head **184** surrounding the first opening **210**. The first member **181** may also include a first position indicator **213**, such as a series of rule lines that are included on the lip **211** on both sides of the opening **210** and that are evenly spaced and arranged along the first receiver axis **193**.

[0036] The first member **181** may also include one or more first engagement features **220** (FIGS. **3** and **5**) for engaging and retaining the second member **182** in a fixed position. There may be a plurality of first engagement features **220**. The first engagement features **220** may be configured to correspond and compliment the shape, dimension, etc. of similar engagement features of the second member **182**. In some embodiments, the plurality of engagement features **220** may comprise an array of ridges **222** and respective slots **224** defined between neighboring pairs of ridges **222** (FIG. **5**). The ridges **222** and slots **224** may be included within the first opening **210**, on the interior surfaces of the first wall **201** and third wall **203**, and the ridges **222** and slots **224** may extend longitudinally along the second receiver axis **195**. In other words, an insertion axis of the slots **224** may extend parallel to the receiver axis **195**. The first member **181** may include any suitable number of ridges **222** and slots **224** therebetween. For example, in some embodiments, there may be fourteen ridges **222** on the first wall **201** and fourteen corresponding ridges **222** on the third wall **203**. Conversely there may be fifteen slots **224** on the first wall **201** and fifteen corresponding slots **224** on the third wall **203**. The slots **224** and ridges **222** may be aligned across the first receiver axis **193**. The ridges **222** may have a tapered cross section (i.e., may be individually tapered in a cross-section taken normal to the second receiver axis **195** as shown in FIG. **5**) so as to taper down to a point on the interior ends of the ridges **222**. The slots **224** may be tapered inversely as compared to the ridges **222**. Thus, the slots **224** may taper down to a point within the slots **224**. Accordingly, the first engagement features **220** may have a sawtooth-shaped cross-sectional profile (FIG. **5**). In some embodiments, a majority (e.g., substantially all) of the first wall **201** and the third wall **203** may be ribbed with the first engagement features **220**. The remainder of the first opening **210** may be defined by the substantially flat interior surfaces of the second wall **202** and the fourth wall **204**.

[0037] Furthermore, the first member **181** may include at least one snap-fit projection **230**. In some embodiments, there may be two snap-fit projections **230**, and one may project from the first wall **201** and the other may project from the third wall **203** (FIG. **7**). The snap-fit projections **230** may be resiliently flexible. In a neutral, unbiased position, one of the projections **230** may be a flat, rectangular flap that projects outward to be cantilevered from the first wall **201** along the third receiver axis **197** and upward toward the underside of the support head **184**. The other projection **230** may be similar, except that it may project outward from the third wall **203** and upward toward the underside of the support head **184**. The projections **230** may resiliently bend or otherwise deflect toward the first wall **201** and third wall **203**, respectively, and may resiliently recover toward the neutral position represented in FIG. **3**. Accordingly, the first member **181** may be conveniently snap-fit into the sill **149** in some embodiments as will be discussed in greater detail below.

[0038] In some embodiments, the first member **181** may be symmetric with respect to at least one line of symmetry. For example, the first member **181** may be symmetric about the first receiver axis **193**. As will be appreciated, this symmetry may make installation convenient because the first member **181** may be robustly attached to the sill **149** in multiple orientations.

[0039] Moreover, as shown in FIG. **7**, the first member **181** may be formed in a molding device **240**. The molding device **240** may include a first mold member **241** and a second molding member **242** that may be fit together to define a mold cavity **244** (e.g., injection molding). Material (e.g., polymeric material) may be provided in the mold cavity **244**, and the mold members **241**, **242** may shape and form the first member **181** according to the interior surfaces of the mold cavity **244**. Accordingly, the first member **181** may be formed efficiently in the molding device **240**.

[0040] Referring now to FIGS. **3**, **8**, and **9**, the second member **182** may be a unitary, one-piece, monolithic part. The second member **182** may be constructed of a rigid and strong polymeric material. In some embodiments, the second member **182** may be constructed of a material suitable for molding (e.g., for injection molding). The second member **182** may include a strike head **246** and an insert projection **248** that projects from the strike head **246** along the second receiver axis **195**.

[0041] The strike head **246** may comprise a flat plate that lies substantially within a plane defined by the first and third receiver axes **193**, **197**. The strike head **246** may include an upper strike surface **247** that faces upward along the second receiver axis **195**. In some embodiments, the upper strike surface **247** may include a second position indicator **249**, such as one or more rule lines that are centered on the surface **247** along the first receiver axis **193**.

[0042] The insert projection **248** may be hollow and tubular and may include a first wall **251**, a second wall **252**, a third wall **253**, and a fourth wall **254** that extend along the second receiver axis **195** from the underside of the strike head **246**. The second member **182** may further include the second opening **250** mentioned above in reference to FIG. **2**. In some embodiments, there may be a large-radius, convex transition between the upper strike surface **247** and the interior side of the second wall **252**, and a similar transition between the upper strike surface **247** and the interior side of the fourth wall **254**.

[0043] The second opening **250** may be a through-hole that extends through the strike head **246** and that is defined between the walls **251**, **252**, **253**, **254** of the insert projection **248**. The second opening **250** may be rectangular (e.g., substantially square) in a cross-section taken normal to the second receiver axis **195**. The first receiver axis **193** may extend between the second and fourth walls **252**, **254**, and the third receiver axis **197** may extend between the first and third walls **251**, **253**.

[0044] The second member **182** may also include at least one second engagement feature **258** for engaging with the first member **181** in a plurality of different fixed positions. The second engagement feature **258** may correspond in shape, position, etc. to the first engagement features **220** of the first member **181** for selectively engaging therewith. In some embodiments, the second engagement feature **258** may include one or more flanges **260** extending from the insert projection **248**. Specifically, there may be four flanges **260** (i.e., a first, second, third, and fourth flange **260**) that extend outwardly from the respective corners of the insert projection **248** along the third receiver axis **197**. The flanges **260** may be tapered and may taper down to a point as the flanges **260** extend along the third receiver axis **197**. The flanges **260** may be aligned across the first receiver axis **193**.

[0045] In some embodiments, the second member **182** may be symmetric with respect to at least one line of symmetry. For example, the second member **182** may be symmetric about the first receiver axis **193**. As will be appreciated, this symmetry may make installation convenient because the second member **182** may be robustly attached to the first member **181** in multiple orientations.

[0046] Moreover, as shown in FIG. **9**, the second member **182** may be formed in a molding device **262**. The molding device **262** may include a first mold member **264** and a second molding member **266** that may be fit together to define a mold cavity **268** (e.g., for injection molding). Material (e.g., polymeric material) may be provided in the mold cavity **268**, and the mold members **264**, **266** may shape and form the second member **182** according to the interior surfaces of the mold cavity **268**. Accordingly, the second member **182** may be formed efficiently in the molding device **262**.

[0047] Referring now to FIGS. **2-4**, installation and use of the adjustable bolt receiver **170** and the bolt securement system **160** will be discussed according to example embodiments. The bolt **162** may be slidably mounted (e.g., with fasteners) to the panel **108** and selectively moved to the retracted position (shown in phantom in FIG. **2**). Then, the first member **181** may be attached to the sill **149**. For example, the sill aperture **290** may be formed (e.g., by drilling) into the top wall **172**. The sill aperture **290** may be positioned into rough alignment with the bolt **162**.

[0048] Then, the first member **181** may be manually pushed into a sill aperture **290** in the top wall **172**. The sill aperture **290** may be slightly larger than the periphery of the base extension **186**, and as the first member **181** is advanced along the second receiver axis **195**, the snap-fit projections **230** may resiliently bend inward to clear the rim of the sill aperture **290**. Once the snap-fit projections **230** have advanced beyond the sill aperture **290** and into the sill cavity **152**, the projections **230** may resiliently recover, thereby securing the first member **181** to the sill **149**.

[0049] Next, a position for the second member **182** within the first member **181** may be determined and selected. More specifically, the user may move the panel **108** to a desired position within the frame **104** (e.g., to a closed position), and the bolt **162** may be moved to its extended position (shown in solid lines in FIG. **2**) and into the first opening **210**. Accordingly, the user may use the first position indicator **213** to visually confirm where the sliding axis of the bolt **162** (axis **105**) is located along the first receiver axis **193**. With this information, the user may manually insert (e.g., push) the insert projection **248** into the first opening **210** until the strike head **246** overlaps and abuts the lip **211** of the support head **184**. The flanges **260** of the second member **182** may be snugly fit, may nest, or may otherwise be received in respective ones of the slots **224** of the first member **181**. As such, the second member **182** may be robustly retained in a fixed position along the first receiver axis **193** due to interference between the flanges **260** and the ridges **222**. If properly positioned, the second position indicator **249** will align with the line of the first position indicator **213** that was previously identified as being aligned with the bolt **162**. As such, the second opening **250** may be conveniently positioned and aligned along the first receiver axis **193** with the bolt **162**.

[0050] If further adjustment is needed, then the second member **182** may be manually pulled and removed from the first member **181**, and the second member **182** may be re-inserted at the adjusted position along the first receiver axis **193** with the flanges **260** fitting with other ones of the slots **224** of the first member **181**. FIG. **4** illustrates two of a plurality of potential positions of the second member **182** within the first member **181**. Specifically, a first, centered position of the second member **182** is shown in solid lines, and a second position of the second member **182** is shown in phantom. It will be appreciated that the second member **182** may be fit into additional positions within the first member **181** as well. In some embodiments, the adjustable bolt receiver **170** may allow for up to ⅜ inches of adjustment (i.e., 0.375 inches or 9.5 mm of adjustment) relative to the first member **181** along the first receiver axis **193**.

[0051] Thus, the adjustable bolt receiver **170** may make alignment of the bolt and the bolt receiver to be highly convenient and less time consuming. The receiver **170** may be strong and robust for securing the panel **108** to the frame **104**. Also, the adjustable bolt receiver **170** may have a relatively low part count for added manufacturing efficiency, reduction of costs, ease of assembly, etc. The adjustable bolt receiver **170** may be assembled and robustly retained without the need for fasteners (i.e., a fastener-less assembly). The adjustable bolt receiver **170** may be highly manufacturable and may be packaged neatly. Moreover, the bolt receiver **170** may also be compact, may have a low-profile, and may be aesthetically pleasing.

[0052] Additional example embodiments of the present disclosure may include an adjustable bolt receiver for selectively securing a panel of a fenestration unit to a panel support of the fenestration unit is disclosed wherein the fenestration unit has a bolt that is slidably attached to the panel for movement between a retracted position and an extended position relative to the panel for selectively engaging with the panel support. The adjustable bolt receiver includes a first member configured to be fixedly attached to the panel support, and the first member has a first opening defining an axis. The first member includes at least one first engagement feature. Moreover, the adjustable bolt receiver includes a second member that includes a strike head and an insert projection that projects from the strike head. The insert projection is receivable in the first opening in a first position along the axis and, alternatively, in a second position along the axis. The second member includes at least one second engagement feature configured to engage with the at least one first engagement feature for retaining the insert projection in the first position and in the second position along the axis. The insert projection at least partly defines a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support.

[0053] In some embodiments, the second opening is a through-hole extending along an insertion axis through the strike head and the insert projection.

[0054] Also, in some embodiments, the second opening is a rectangular through-hole.

[0055] In some embodiments, the first member includes a lip, and the strike head overlaps the lip in the first position and in the second position.

[0056] In some embodiments, the lip includes a first position indicator and the strike head includes a second position indicator that aligns with the first position indicator.

[0057] Moreover, in some embodiments the first member is symmetrical about the axis.

[0058] Furthermore, in some embodiments, the second member is symmetrical about the axis.

[0059] In some embodiments, both the first member and the second member are symmetrical about the axis.

[0060] Also, in some embodiments, the at least one first engagement feature comprises an array of slots, and the at least one second engagement feature includes at least one flange extending from the insert projection.

[0061] In some embodiments, the at least one flange includes a first flange and a second flange disposed on opposite sides of the axis.

[0062] Additionally, in some embodiments, individual ones of slots in the array of slots may extend longitudinally along an insertion axis and may be individually tapered in a cross-section taken normal to the insertion axis. The at least one flange is tapered in the cross-section.

[0063] Moreover, in some embodiments, the first member includes a support head and a base extension that extends from the support head. The base extension includes the first opening, the base extension includes the plurality of first engagement features, and the base extension is insertable into the panel support.

[0064] In some embodiments, the base extension includes a snap-fit projection that is resiliently flexible for snap-fitting into the panel support.

[0065] In some embodiments, the first member and the second member are constructed of a polymeric material.

[0066] Also, in another embodiment, a method of manufacturing an adjustable bolt receiver configured for selectively securing a panel of a fenestration unit to a panel support of the fenestration unit is disclosed, wherein the fenestration unit has a bolt that is slidably attached to the panel for movement between a retracted position and an extended position relative to the panel for selectively engaging with the panel support. The method includes providing a first member configured to be fixedly attached to the panel support, and the first member has a first opening defining an axis. The first member includes at least one first engagement feature. Furthermore, the method includes providing a second member that includes a strike head and an insert projection that projects from the strike head. The second member includes at least one second engagement feature, and the insert projection at least partly defining a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support. The method also includes selecting one of a first position for the insert projection in the first opening along the axis and a second position for the insert projection in the second opening along the axis. Additionally, the method includes inserting the insert projection in the selected one of the first position and the second position. The at least one second engagement feature is configured to engage with the at least one first engagement feature for retaining the insert projection in the first position and, alternatively, in the second position along the axis.

[0067] In some embodiments, the method includes providing the first member includes molding the first member.

[0068] Also, in some embodiments, the method includes providing the second member includes molding the second member.

[0069] Additionally, in a further embodiment, a fenestration unit is disclosed that includes a panel with a bolt that is slidably attached thereto for movement between a retracted position an extended position relative to the panel. The fenestration unit includes a frame that supports the panel for sliding movement within the frame along an axis, and the frame includes a sill. Moreover, the fenestration unit includes an adjustable bolt receiver that includes a first member configured to be fixedly attached to the sill, and the first member has a first opening that extends along the axis. The first member includes at least one array of ridges and slots. Also, the adjustable bolt receiver includes a second member that includes a strike head and an insert projection that projects from the strike head. The insert projection is receivable in the first opening in a first position along the axis and, alternatively, in a second position along the axis. The second member includes at least one flange that is removably engaged within one of the slots of the array in the first position and in another of the slots of the array in the second position. The insert projection at least partly defines a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support.

[0070] In some embodiments, the first member and the second member are symmetrical about the axis.

[0071] Also, in some embodiments, the first member includes a snap-fit projection that is resiliently flexible for snap-fitting into the sill.

[0072] It is intended that subject matter disclosed in particular portions herein can be combined with the subject matter of one or more of other portions herein as long as such combinations are not mutually exclusive or inoperable. In addition, many variations, enhancements and modifications are possible.

[0073] While at least one exemplary embodiment has been presented in the foregoing detailed description of the disclosure, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing an exemplary embodiment of the disclosure. It is understood that various changes may be made in the function and arrangement of elements described in an exemplary embodiment without departing from the scope of the disclosure as set forth in the appended claims.

What is claimed is:

1. An adjustable bolt receiver for selectively securing a panel of a fenestration unit to a panel support of the fenestration unit, the fenestration unit having a bolt that is slidably attached to the panel for movement between a retracted position and an extended position relative to the panel for selectively engaging with the panel support, the adjustable bolt receiver comprising:

a first member configured to be fixedly attached to the panel support, the first member having a first opening defining an axis, the first member including at least one first engagement feature; and

a second member that includes a strike head and an insert projection that projects from the strike head, the insert projection being receivable in the first opening in a first position along the axis and, alternatively, in a second position along the axis, the second member including at least one second engagement feature configured to engage with the at least one first engagement feature for retaining the insert projection in the first position and in the second position along the axis, the insert projection at least partly defining a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support.

2. The adjustable bolt receiver of claim **1**, wherein the second opening is a through-hole extending along an insertion axis through the strike head and the insert projection.

3. The adjustable bolt receiver of claim **2**, wherein the second opening is a rectangular through-hole.

4. The adjustable bolt receiver of claim **1**, wherein the first member includes a lip, wherein the strike head overlaps the lip in the first position and in the second position.

5. The adjustable bolt receiver of claim **4**, wherein the lip includes a first position indicator and the strike head includes a second position indicator that aligns with the first position indicator.

6. The adjustable bolt receiver of claim **1**, wherein the first member is symmetrical about the axis.

7. The adjustable bolt receiver of claim **1**, wherein the second member is symmetrical about the axis.

8. The adjustable bolt receiver of claim **1**, wherein both the first member and the second member are symmetrical about the axis.

9. The adjustable bolt receiver of claim **1**, wherein the at least one first engagement feature comprises an array of slots, and wherein the at least one second engagement feature includes at least one flange extending from the insert projection.

10. The adjustable bolt receiver of claim **9**, wherein the at least one flange includes a first flange and a second flange disposed on opposite sides of the axis.

11. The adjustable bolt receiver of claim **9**, wherein individual ones of slots in the array of slots extend longitudinally along an insertion axis and are individually tapered in a cross-section taken normal to the insertion axis; and

wherein the at least one flange is tapered in the cross-section.

12. The adjustable bolt receiver of claim **1**, wherein the first member includes a support head and a base extension that extends from the support head, the base extension including the first opening, the base extension including the plurality of first engagement features, the base extension being insertable into the panel support.

13. The adjustable bolt receiver of claim **12**, wherein the base extension includes a snap-fit projection that is resiliently flexible for snap-fitting into the panel support.

14. The adjustable bolt receiver of claim **1**, wherein the first member and the second member are constructed of a polymeric material.

15. A method of manufacturing an adjustable bolt receiver configured for selectively securing a panel of a fenestration unit to a panel support of the fenestration unit, the fenestration unit having a bolt that is slidably attached to the panel for movement between a retracted position and an extended position relative to the panel for selectively engaging with the panel support, the method comprising:

providing a first member configured to be fixedly attached to the panel support, the first member having a first opening defining an axis, the first member including at least one first engagement feature;

providing a second member that includes a strike head and an insert projection that projects from the strike head, the second member including at least one second engagement feature, the insert projection at least partly defining a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support;

selecting one of a first position for the insert projection in the first opening along the axis and a second position for the insert projection in the second opening along the axis; and

inserting the insert projection in the selected one of the first position and the second position, the at least one second engagement feature configured to engage with the at least one first engagement feature for retaining the insert projection in the first position and, alternatively, in the second position along the axis.

16. The method of claim **15**, wherein providing the first member includes molding the first member.

17. The method of claim **15**, wherein providing the second member includes molding the second member.

18. A fenestration unit comprising:

a panel with a bolt that is slidably attached thereto for movement between a retracted position an extended position relative to the panel;

a frame that supports the panel for sliding movement within the frame along an axis, the frame including a sill; and

an adjustable bolt receiver that includes:

a first member configured to be fixedly attached to the sill, the first member having a first opening that extends along the axis, the first member including at least one array of ridges and slots; and

a second member that includes a strike head and an insert projection that projects from the strike head, the insert projection being receivable in the first opening in a first position along the axis and, alternatively, in a second position along the axis, the second member including at least one flange that is removably engaged within one of the slots of the array in the first position and in another of the slots of the array in the second position, the insert projection at least partly defining a second opening configured to receive the bolt when in the extended position to secure the panel to the panel support.

19. The fenestration unit of claim **18**, wherein the first member and the second member are symmetrical about the axis.

20. The fenestration unit of claim **18**, wherein the first member includes a snap-fit projection that is resiliently flexible for snap-fitting into the sill.

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